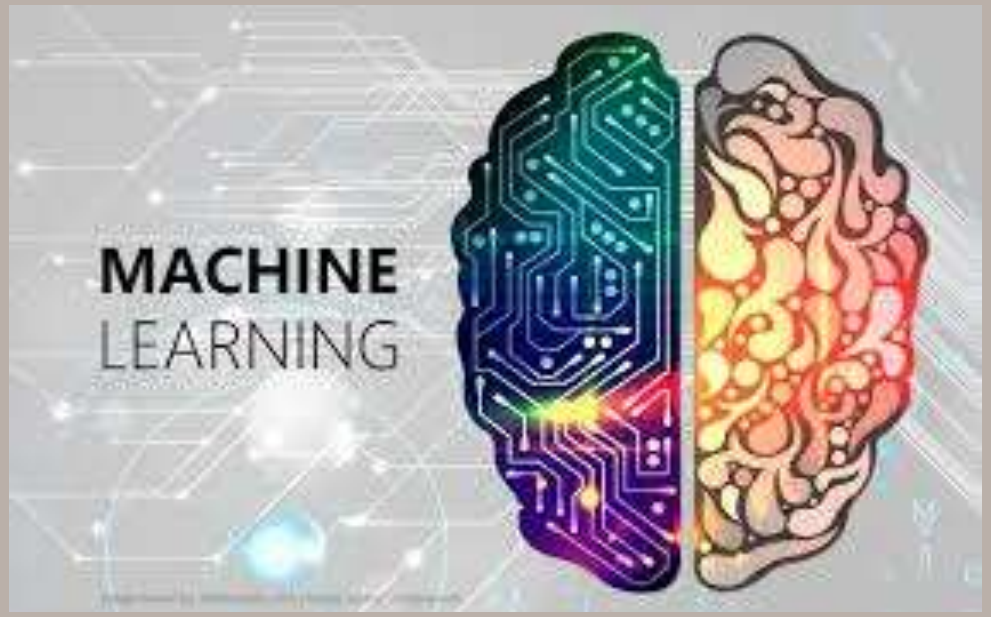
CSE 4137

Machine Learning
Module-01

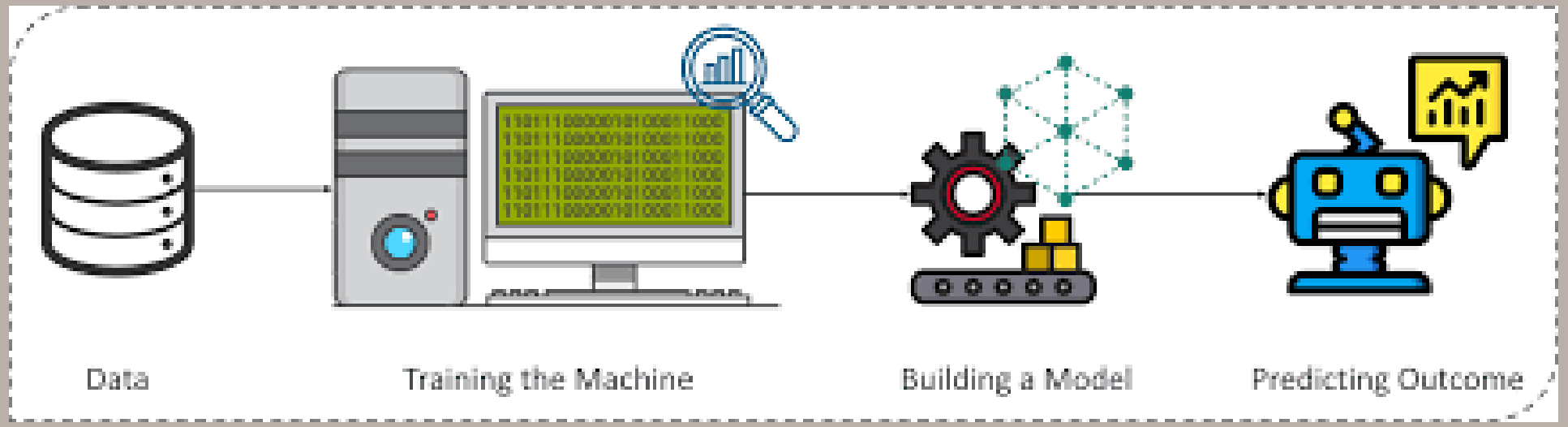


Md. Shahid Uz Zaman
Adjunct Faculty
Dept of CSE, BAUET
szaman22.ruet@gmail.com

Topics

Week-01

- Introduction to ML
- History of ML
- Types of ML



What is Machine Learning?

Introduction to ML

W-01, C-01

What is Learning (ML)?

"Learning is any process by which a system improves performance from experience." - Herbert Simon

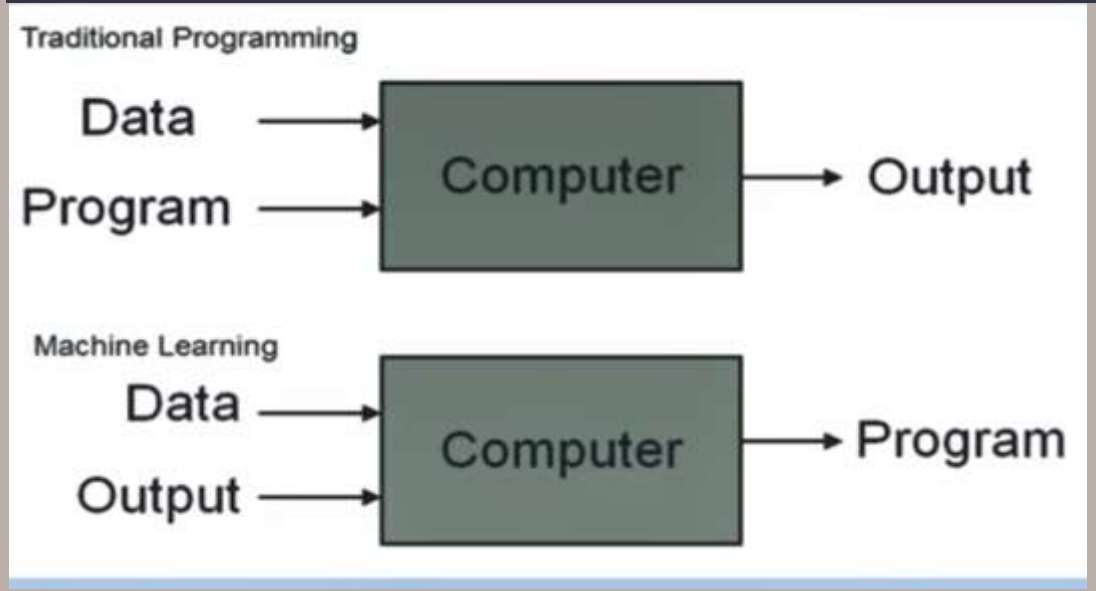
What is Explicit Programming?

Writing code for every scenario. In ML we don't do that. In ML logic is generated based on input and output data

What is Machine Learning (ML)?

Machine learning is a field of computer science that uses statistical techniques to give computer systems the ability to "learn" with data, without being explicitly programmed.

Traditional Programming vs ML



Simply, ML is "Machine learns from data"

Introduction to ML

W-01, C-01

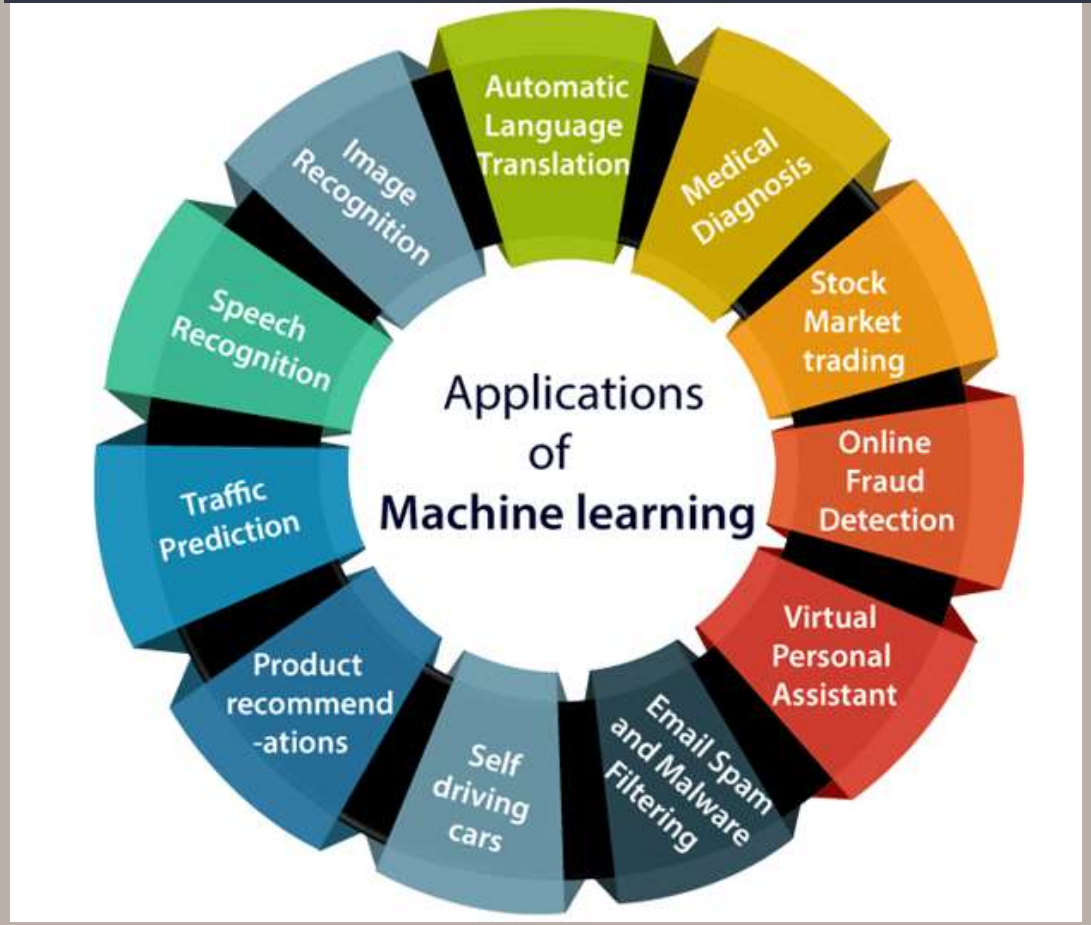
Example: ML

Suppose to add two numbers we can write a program (sum=a+b). The program can explicitly add two numbers only. It can't add more than three numbers. But is ML data (like excel sheet) input and output is provided and Machine learns the pattern that it is plus/addition. Therefore ML can add more than two numbers (any numbers).. That's why ML is so powerful now-a-days.

The Scenario where ML is used

Sometimes you can not write program to handle all the situations.. Ex: Email classifier, Image classifier etc.

Applications: ML

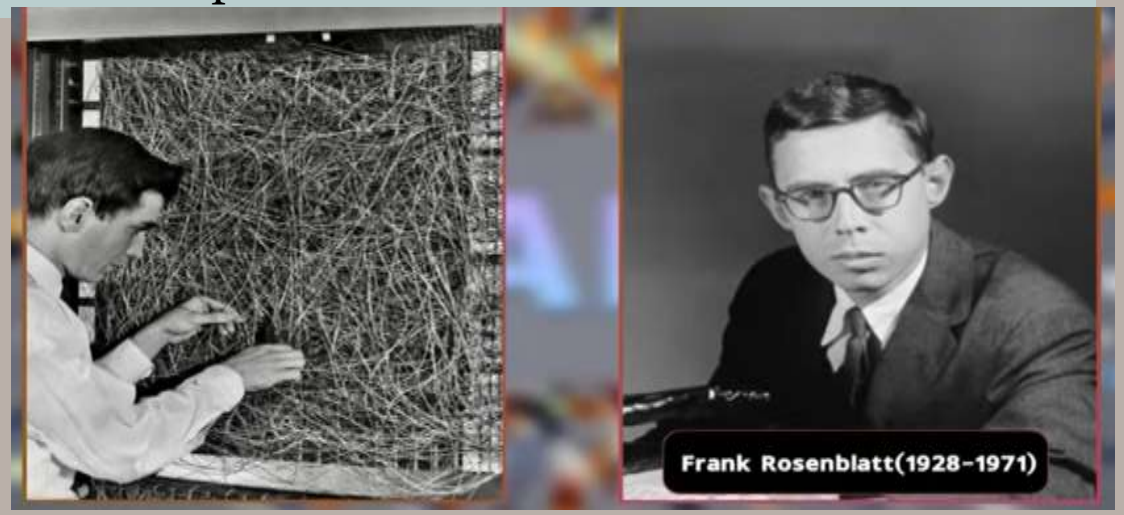


History of ML

W-01, C-01

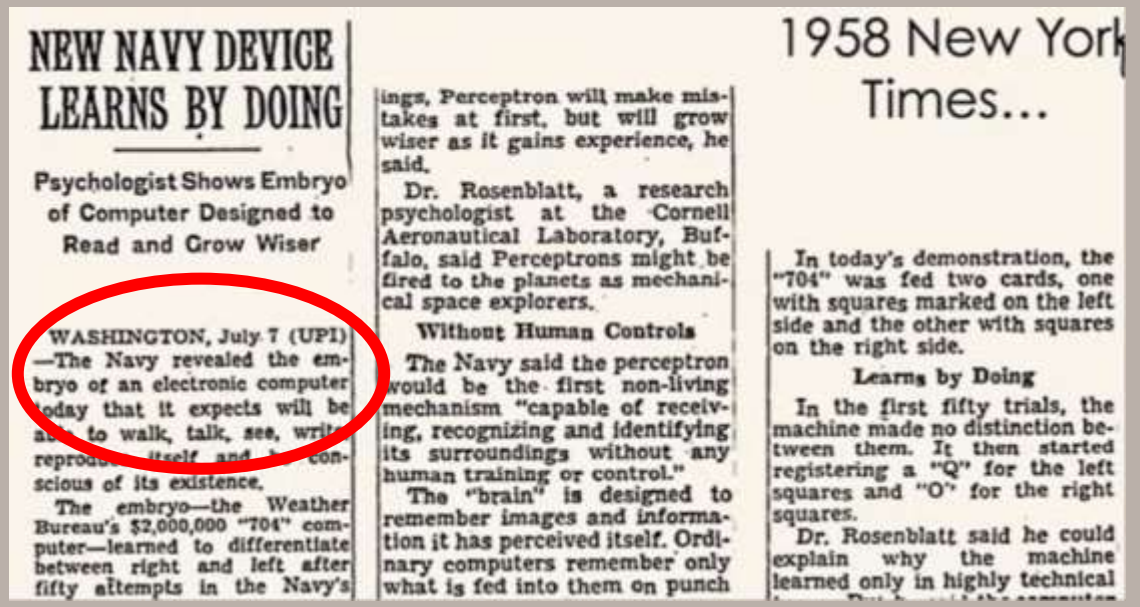
History of ML

The history of ML is very old. In 1957 Psychologist Frank Rosenblatt made a Neural Network that can identify male and female picture. ML was unable to progress due to lack of data and powerful hardware.









NEW NAVY DEVICE LEARNS BY DOING

Psychologist Shows Embryo of Computer Designed to Read and Grow Wiser

WASHINGTON, July 7 (UPI) —The Navy revealed the embryo of an electronic computer today that it expects will be able to walk, talk, see, write, reproduce itself and be conscious of its existence.

The embryo—the Weather Bureau's $2,000,000 "704" computer—learned to differentiate between right and left after fifty attempts in the Navy's ings, Perceptron will make mistakes at first, but will grow wiser as it gains experience, he said.

Dr. Rosenblatt, a research psychologist at the Cornell Aeronautical Laboratory, Buffalo, said Perceptrons might be fired to the planets as mechanical space explorers.

Without Human Controls

The Navy said the perceptron would be the first non-living mechanism "capable of receiving, recognizing and identifying its surroundings without any human training or control."

The "brain" is designed to remember images and information it has perceived itself. Ordinary computers remember only what is fed into them on punch

1958 New York Times...

In today's demonstration, the "704" was fed two cards, one with squares marked on the left side and the other with squares on the right side.

Learns by Doing

In the first fifty trials, the machine made no distinction between them. It then started registering a "Q" for the left squares and "O" for the right squares.

Dr. Rosenblatt said he could explain why the machine learned only in highly technical



Identify Hand-writing In 1984



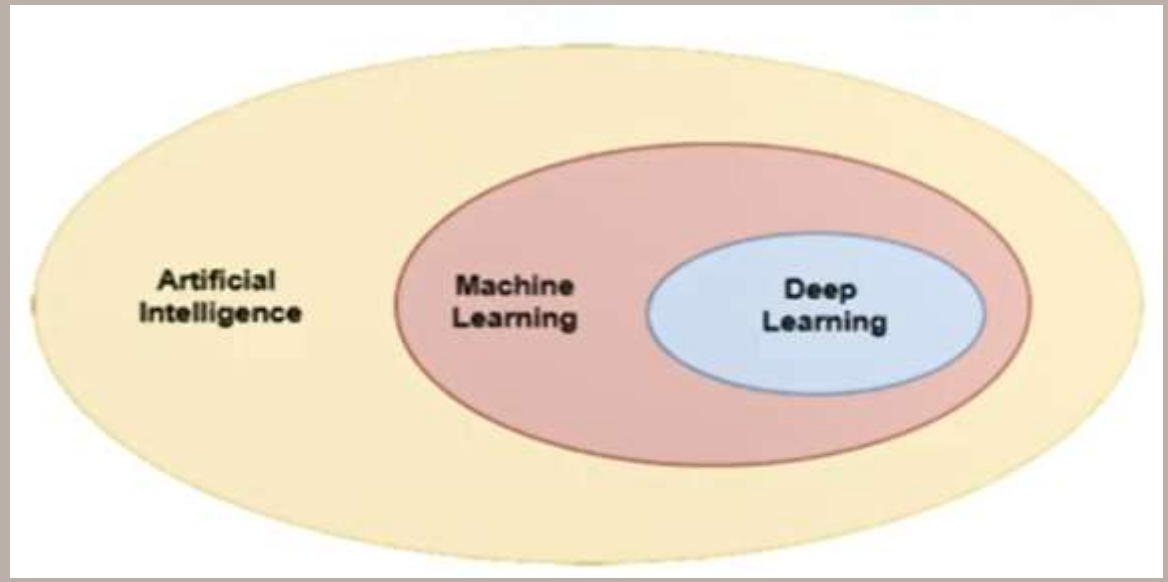
Sefl-driven car in 1986

AI vs ML vs DL

W-01, C-01

AI vs ML vs DL

- **AI** stands for Artificial Intelligence, and is basically the study/process which enables machines to mimic human behavior through particular algorithm.
- **ML** stands for Machine Learning, and is the study that uses statistical methods enabling machines to improve with experience.
- **DL** stands for Deep Learning, and is the study that makes use of Neural Networks(similar to neurons present in human brain) to imitate functionality just like a human brain.



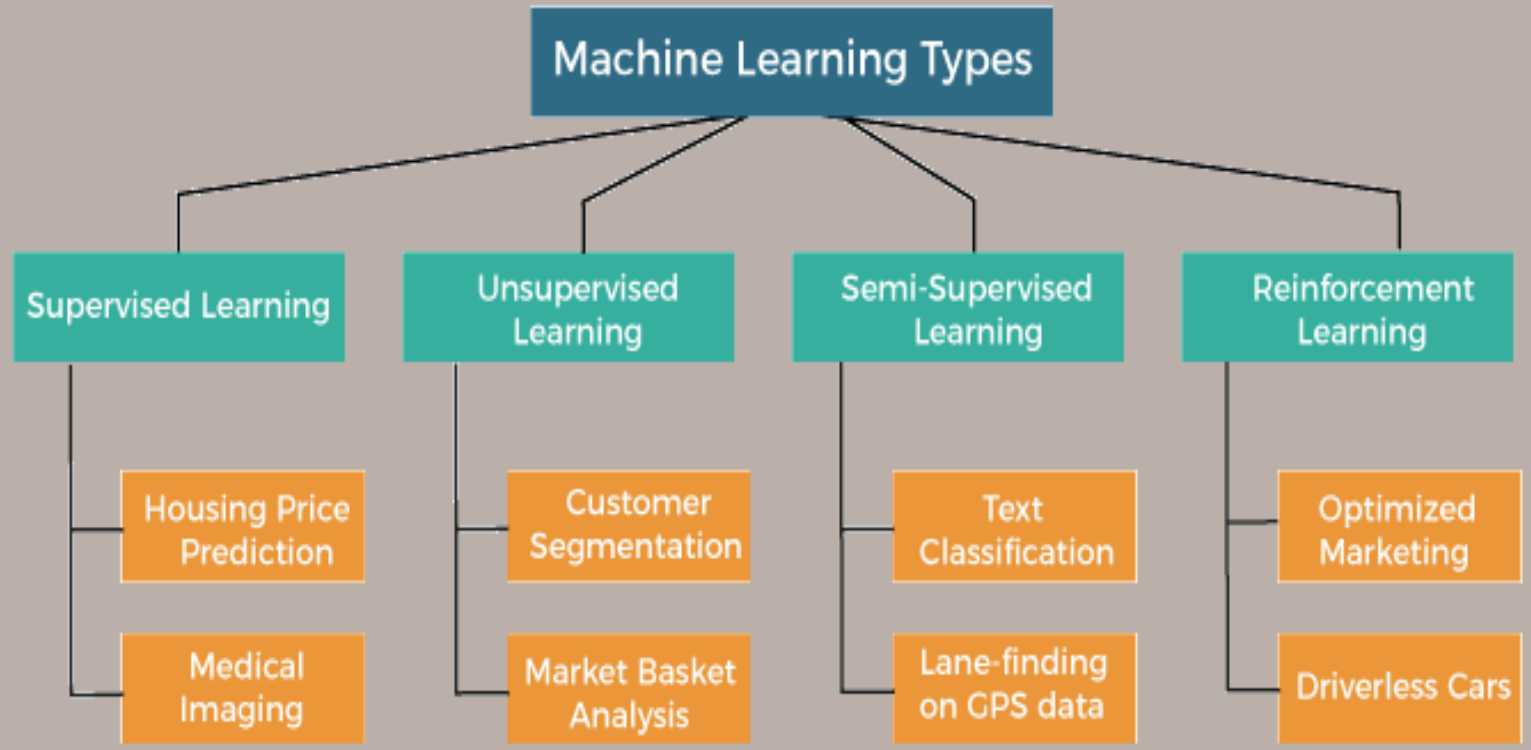
Types of Machine Learning?

Types of ML

W-01, C-02

Types of ML

Based on Supervision, MLs are four types
1. Supervised Machine Learning
2. Unsupervised Machine Learning
3. Semi-Supervised Machine Learning
4. Reinforcement Learning



Types of ML

W-01, C-02

1. Supervised Machine Learning

As its name suggests, **Supervised machine learning** is based on supervision. It means in the supervised learning technique, we train the machines using the "labelled" dataset, and based on the training, the machine predicts the output. Here, the labelled data specifies that some of the inputs are already mapped to the output. More preciously, we can say; first, we train the machine with the input and corresponding output, and then we ask the machine to predict the output using the test dataset.

Let's understand supervised learning with an example. Suppose we have an input dataset of cats and dog images. So, first, we will provide the training to the machine to understand the images, such as the shape & size of the tail of cat and dog, Shape of eyes, colour, height (dogs are taller, cats are smaller), etc. After completion of training, we input the picture of a cat and ask the machine to identify the object and predict the output. Now, the machine is well trained, so it will check all the features of the object, such as height, shape, colour, eyes, ears, tail, etc., and find that it's a cat. So, it will put it in the Cat category. This is the process of how the machine identifies the objects in Supervised Learning.

Example: Supervised ML

Let we have the following table of data. Based on IQ and CGPA, student gets placement.

IQ	CGPA	Placement
7.5	3.5	Yes
6.8	3.8	Yes
5.7	3.1	No
7.3	2.8	No

From the above data machine find a formula between input and output. Use the formula later to predict the placement with new input data. Ex: for IQ=6.2 and CGPA=3.0 it can predict the output although these new data was not present in the table.

Types of ML

W-01, C-02

Types of Supervised ML

a) **Classification**

Classification algorithms are used to solve the classification problems in which the output variable is categorical, such as "**Yes" or No, Male or Female, Red or Blue, etc**. The classification algorithms predict the categories present in the dataset. Some real-world examples of classification algorithms are **Spam Detection, Email filtering, etc. The output should be a categorical data**. (same as the previous example)

b) Regression

Regression algorithms are used to solve regression problems in which there is a linear relationship between input and output variables. These are used to predict continuous output variables, such as market trends, weather prediction, etc.. **The output should be a numeric value**.

Example: Regression SML

Let we have the following table of data. Based on IQ and CGPA, student gets package.

IQ	CGPA	Package (lacs)
7.5	3.5	6
6.8	3.8	5.5
5.7	3.1	4
7.3	2.8	3

After the machine gets trained it can predict the package (output) of any new input (IQ, CGPA).

Types of ML

W-01, C-02

2. Unsupervised ML

Unsupervised learning is different from the Supervised learning technique; as its name suggests, there is no need for supervision. It means, in unsupervised machine learning, the machine is trained using the unlabeled dataset, and the machine predicts the output without any supervision. It has no output data.

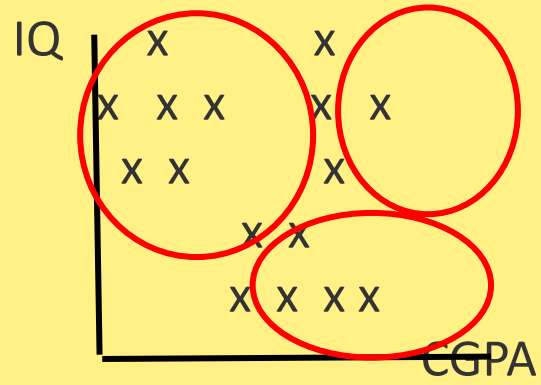
IQ	CGPA	Output
7.5	3.5	
6.8	3.8	
5.7	3.1	
7.3	2.8	

The [illegible]arning algorithm is to group or categories the unsorted dataset according to the similarities, patterns, and differences.

What to do with Unsupervised ML

1. Clustering
2. Dimensionality Reduction
3. Anomaly Detection
4. Association Rule Learning

1. **Clustering**: Data can be divided into categories. Look at the 2D graph below (IQ in Y-axis and CGPA in X-axis)



Here students data is divided into three categories i) High IQ with Low CGPA, ii) High IQ with High CGPA and iii) Low IQ with High CGPA

Introduction to ML

W-01, C-01

Cont:

2 Dimensionality Reduction

Dimensionality reduction is a technique used to reduce the number of features in a dataset while retaining as much of the important information as possible. In other words, it is a process of transforming high-dimensional data into a lower-dimensional space that still preserves the essence of the original data.

For example, image and text have many input columns and it slows the learning process So we need to reduce the input columns keeping

Example: To predict the price of a house, input fields No. of Rooms and No. of Bathrooms can be combined together as Area of the house. Thus 2 input fields converted to 1.

No of Rooms	No. of Bathrooms
4	2



Area
1450

There is a powerful algorithm called PCA on Dimensionality Reduction.

It is know that for multi-dimensional data we can't plot the data (max 3D). Using Dimension Reduction algorithm we can reduce the dimension so that it can be plotted in 3D.

Types of ML

W-01, C-02

3. Anomaly Detection

Anomaly detection, or **outlier detection**, is the identification of observations, events or data points that deviate from what is usual, standard or expected, making them inconsistent with the rest of a data set.
Example: manufacturing defects, credit card processing defect, loan processing defects etc.

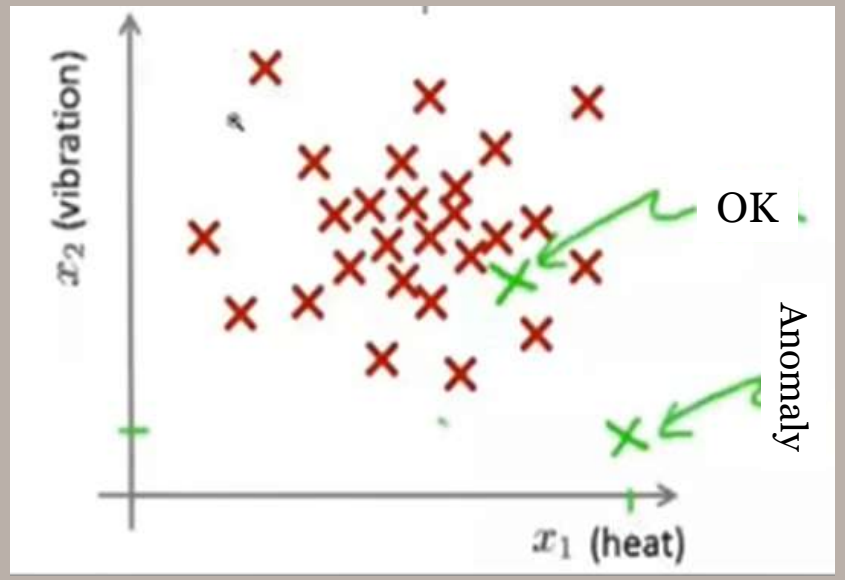
Example: Engine Problem

Aircraft engine features:
- x_1 = heat generated
- x_2 = vibration intensity
- ...

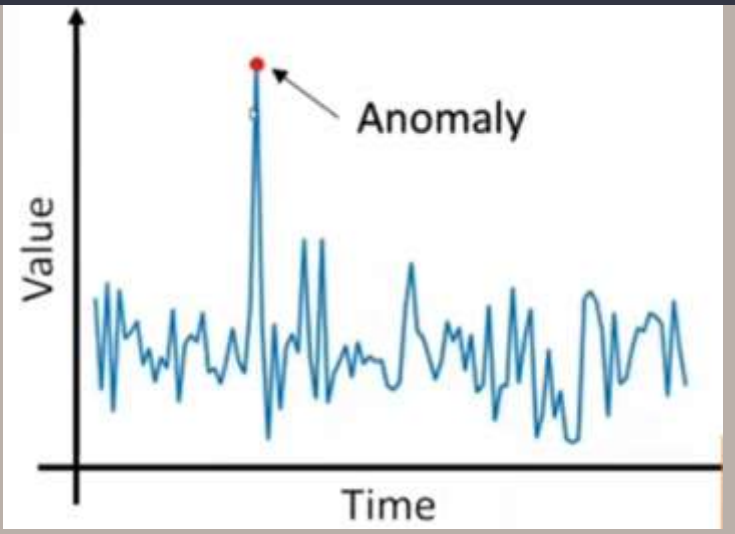
Dataset: $\{x^{(1)}, x^{(2)}, \ldots, x^{(m)}\}$

New engine: x_{test}

Classification Problem



Example: Share Price



Regression Problem

Types of ML

4. Association Rule Based

Association rule learning is an unsupervised learning technique, which finds interesting relations among variables within a large dataset. The main aim of this learning algorithm is to find the dependency of one data item on another data item and map those variables accordingly so that it can generate maximum profit. This algorithm is mainly applied in Market Basket analysis, Web usage mining, continuous production, etc.

For example: Suppose if we check the bills of a departmental store then we observe that 80% customer bought milk and 60% of those also bought egg. So there is a relation/association between Milk and Egg. So in the store they can be arranged aside.

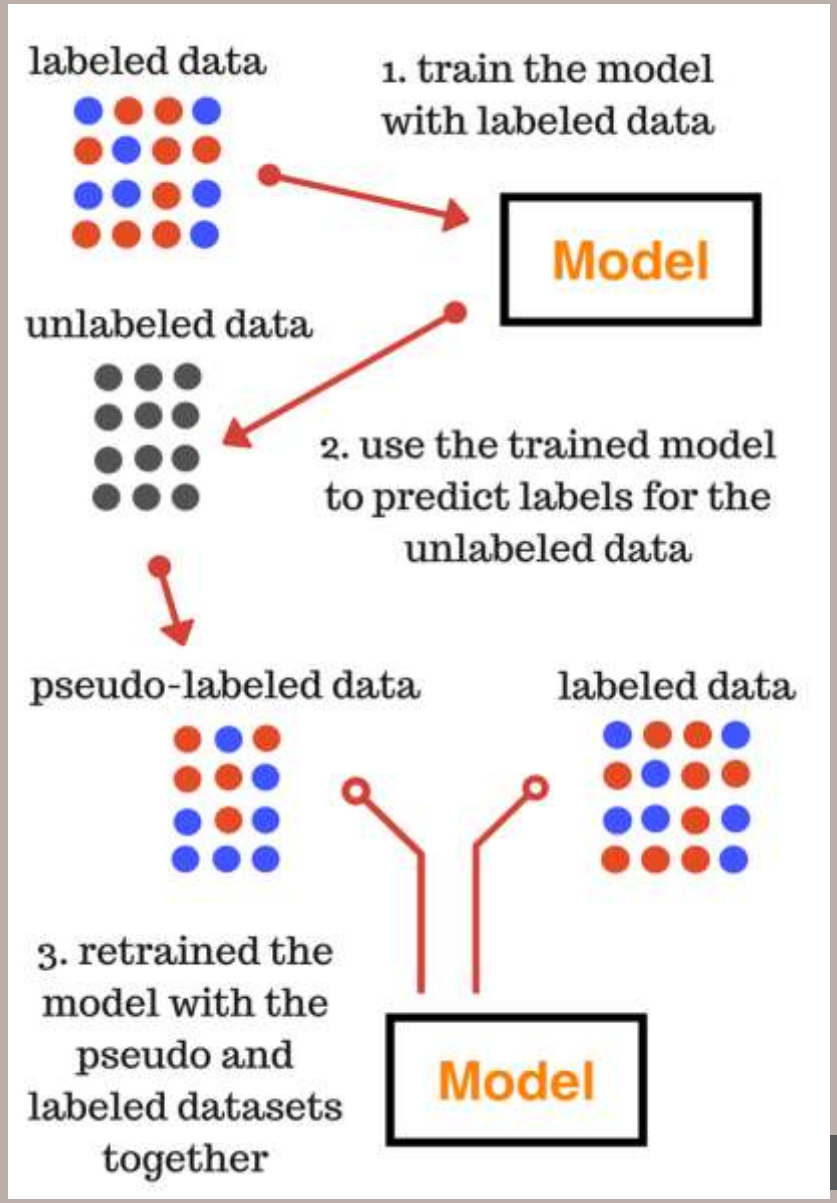


Types of ML

3. Semi-Supervised Learning

Semi-Supervised learning is a type of Machine Learning algorithm that lies between Supervised and Unsupervised machine learning. It represents the intermediate ground between Supervised (With Labelled training data) and Unsupervised learning (with no labelled training data) algorithms and uses the combination of labelled and unlabeled datasets during the training period.

Labeled Field: It costly to create the labeled fields. Human efforts are required to label output data. Example, for Dog Cat classification problem we need label images which dog or cat and it is labeled by human. Input data is easy to get, we can manage so many images but labeling is the problem.
Labeling is costly as well as time consuming.

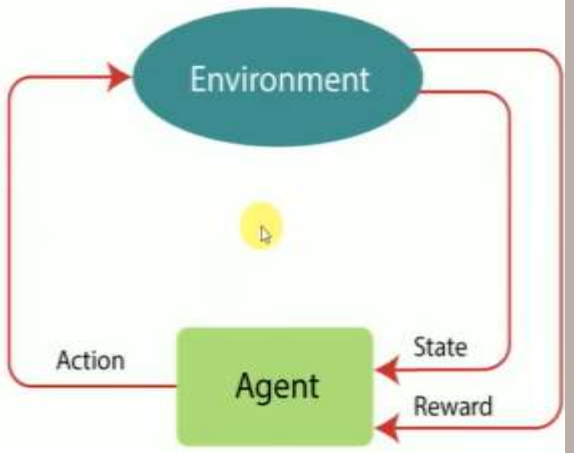


Types of ML

W-01, C-03

4. Reinforcement Learning

- Reinforcement Learning is a feedback-based Machine learning Approach here an agent learns to which actions to perform by looking at the environment and the results of actions.
- For each correct action, the agent gets positive feedback, and for each incorrect action, the agent gets negative feedback or penalty.



How does RL work?

The learning process of reinforcement learning (RL) algorithms is similar to animal and human reinforcement learning in the field of behavioral psychology. For instance, a child may discover that they receive parental praise when they help a sibling or clean but receive negative reactions when they throw toys or yell. Soon, the child learns which combination of activities results in the end reward.

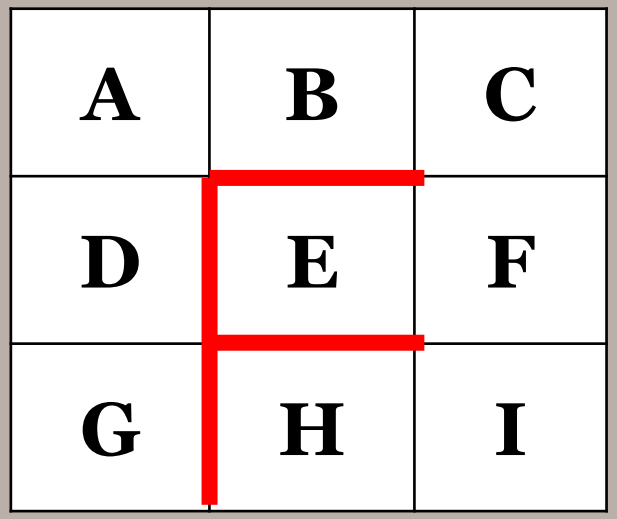
Key Concepts of RL

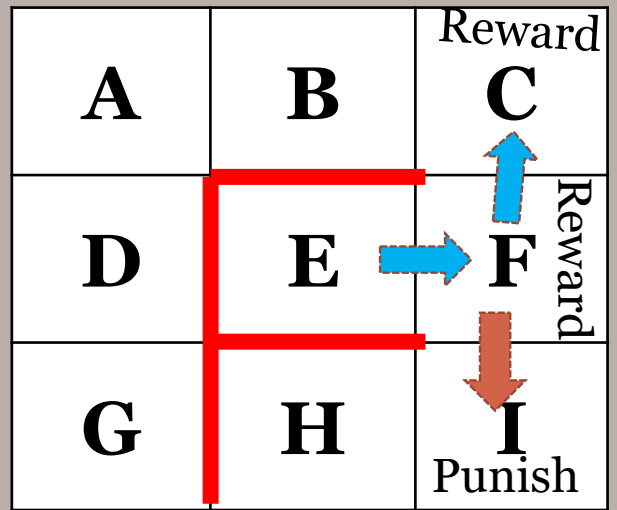
- The **agent** is the ML algorithm (or the autonomous system)
- The **environment** is the adaptive problem space with attributes such as variables, boundary values, rules, and valid actions
- The **action** is a step that the RL agent takes to navigate the environment
- The **state** is the environment at a given point in time
- The **reward** is the positive, negative, or zero value—in other words, the reward or punishment—for taking an action
- The **cumulative reward** is the sum of all rewards or the end value

Types of ML

W-01, C-03

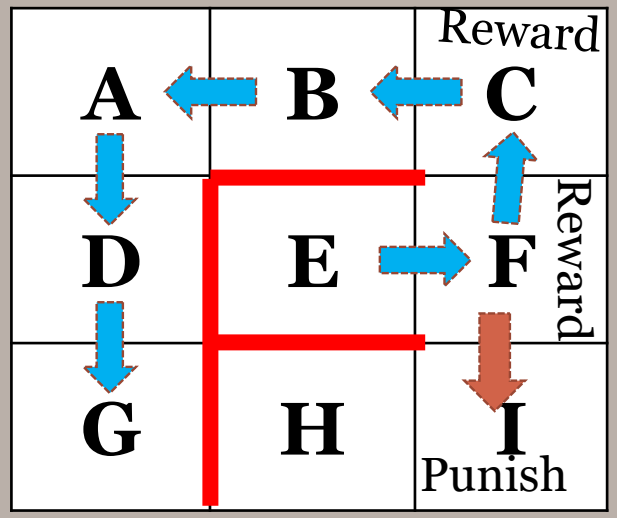
Example of RL





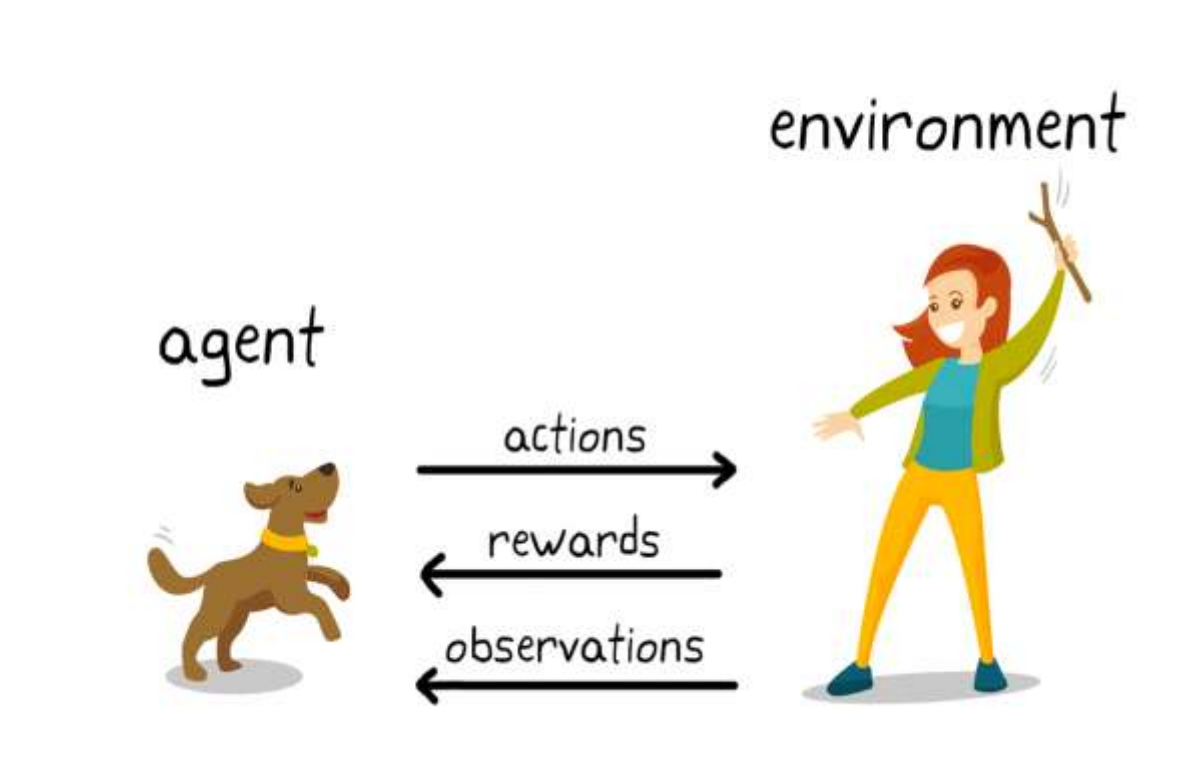
Key Concepts of RL

Here Agent is at block E and Target block is at block G.



Types of ML

Example of RL





W-01, C-03

End